Topological Zero Transport and the Phase of the Discriminant

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Abstract

A multiple zero of an analytic backward heat flow carries a positive local degree. Integrating the normalized first jet over a complete spacetime boundary therefore relates an initial integral to later collision charge. We evaluate the terminal integral on moving root tubes as the phase increment of one scalar function: the squared Vandermonde multiplied by the analytic factor evaluated at the roots. The projective side walls have exactly zero flux. For the canonical de Bruijn–Newman flow we prove uniform complex-time cap estimates for the first $K = o(L\sqrt{\kappa})$ roots, where $L \sim \frac{1}{4}\log T$ and $\kappa \sim 8TL$. The Vandermonde phase error is linear in the population; the analytic factor supplies the leading cap flux. Imaginary-time flux per root is $O(1/\log T)$ on proportional late-time intervals. At a fixed event core, shrinking the terminal caps instead concentrates half the charge on each imaginary-time face. These estimates distinguish the terms that become small from those that retain the topology. The topological and algebraic ingredients are classical; the paper's focus is their quantitative boundary realization, not a new invariant or an RH conclusion.

1 One boundary formula

The useful question is not only where the zeros of a heat-evolved function lie, but how their creation can be measured from a boundary. A boundary formula separates that question into concrete analytic tasks: estimate a terminal cap, choose side walls with manageable flux, and account for the faces needed to close the domain. This paper develops that separation for the actual first jet (H, H_z).

Let $\tau = t + is$ and $z = x + iy$, and suppose that H is holomorphic and satisfies $H_\tau = -H_{zz}$. Away from joint zeros of (H, H_z), put

$$G = (H, H_z), \qquad \mathcal{A} = \frac{\operatorname{Im}(G^* dG)}{|G|^2}, \qquad \mathcal{C} = \frac{\mathcal{A} \wedge d\mathcal{A}}{4\pi^2}. \tag{1.1}$$

Here "physical jet" means that the two components have precisely this relative normalization. Multiplying both by a positive real scalar does not change the normalized jet. Removing a varying common phase, or rescaling only H_z, generally changes individual boundary integrals.

Write $\mathcal{I}_t$ for a forward-oriented real-time cap integral, $\mathcal{J}_\pm$ for the outward integrals on $s = \pm Y$, and $\mathcal{R}$ for the remaining outward spatial and joining faces. The initial outward face is $-\mathcal{I}_0$. When the terminal cap consists of m regular root disks $|H/H_z| \le \epsilon$, the formula to keep in view is

$$\boxed{\mathcal{I}_0 = \frac{k_\epsilon}{2\pi} \operatorname{Im} \int_{T-iY}^{T+iY} d\log(\Delta\mathscr{U}) + \mathcal{J}_+ + \mathcal{J}_- + \mathcal{R} - \mathcal{Q}, \qquad k_\epsilon = \frac{\epsilon^2}{1+\epsilon^2}.} \tag{1.2}$$

Its scalar and integer terms are

$$\Delta = \prod_{i<j}(z_j - z_i)^2, \qquad \mathscr{U} = \prod_{j=1}^{m} U(\tau, z_j(\tau)), \qquad \mathcal{Q} = \sum_{p \in W:\, H(p)=H_z(p)=0} \binom{m_p}{2}, \tag{1.3}$$

where $H = U\prod_j(z - z_j)$ near the selected roots and m_p is the spatial multiplicity at an interior event. The logarithmic integral uses a continuous phase, so it retains every winding. A complete statement of the geometric hypotheses is Theorem 2.4.

Equation (1.2) is an equality of genuine three-dimensional boundary integrals. It does not replace the initial cap by a one-dimensional integral on the real axis. It counts all complex events inside the chosen four-dimensional domain, including the real events at which new real zeros appear. At a real event of order m, the number of new real zeros is $2\lfloor m/2 \rfloor$; its complex charge is $\binom{m}{2}$. These are different quantities. If the initial slice contains a joint zero, its indentation or a local core must be specified and retained. The theorem does not assign a value to an unspecified singular initial cap.

The three main conclusions are as follows. The terminal cap is exactly the phase of $\Delta\mathscr{U}$, with no derivative or volume loss. A fixed projective radius on a moving root tube makes its spatial wall flux zero. Finally, the remaining imaginary-time faces admit quantitative asymptotics: they are small per root on a late interval $[\theta T, T]$, but retain collision charge when attached to a fixed event core. The distinction matters even when only a relative bound is sought.

1.1 The canonical quantitative result

We use the normalization

$$H(\tau, z) = \int_0^\infty e^{\tau u^2}\Phi(u)\cos(zu)\,du, \tag{1.4}$$
$$\Phi(u) = \sum_{n\geq 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}}.$$

Thus $H(0, z) = \xi(\frac{1}{2} + \frac{iz}{2})/8$. Define the large real saddle $L = L_T$ and its curvature by

$$4\pi e^{4L} = 2TL + 9, \qquad \kappa = 16\pi e^{4L} - 2T = 8TL - 2T + 36. \tag{1.5}$$

For fixed $Y > 0$, $0 < r < 1/4$, and an integer $K = K(T)$, set

$$\varepsilon_T = \frac{K + 1 + L^2}{L\sqrt{\kappa}}, \qquad \epsilon = \frac{r}{L}, \qquad K = o(L\sqrt{\kappa}). \tag{1.6}$$

The two different symbols distinguish the analytic error scale ε_T from the physical tube radius ϵ.

Theorem 1.1 (Growing populations and their boundary flux)**.** *For sufficiently large T, the first K positive real zeros of (1.4) continue to distinct simple roots on $T + i[-Y, Y]$, with disjoint projective inverse disks $|H/H_z| \leq r/L$. For any consecutive block of $m \leq K$ of these roots, its forward physical cap satisfies*

$$\frac{\mathcal{I}_{T,m}}{m} = \frac{r^2 Y L^2}{\pi(L^2 + r^2)} + O_Y\left(\frac{r^2\varepsilon_T}{L^2}\right), \tag{1.7}$$

$$\frac{1}{m}\left|\frac{k_\epsilon}{2\pi}\operatorname{Im}\int_{T-iY}^{T+iY} d\log\Delta\right| = O_Y\left(\frac{r^2\varepsilon_T}{L^2}\right). \tag{1.8}$$

All constants are independent of K and m. The analytic-factor contribution is the difference of these two terms; it has leading value r^2Y/π per root.

Fix also $0 < \theta < 1$. The same population has regular root tubes over $[\theta T, T] + i[-Y, Y]$, with the same constant radius r/L_T on each entire tube. Their projective side flux vanishes, and each imaginary-time face satisfies

$$\frac{\mathcal{J}_{\pm}}{m} = -\frac{r^2 Y \log(1/\theta)}{4\pi L_T} + o_{Y,\theta,r}(L_T^{-1}). \tag{1.9}$$

The population is held fixed while integrating on each such interval; it may grow between choices of T.

Theorem 1.1 concerns increasing heat time, with ranks allowed to grow. It is not a high-height theorem at fixed heat time. For example $K = \lfloor \sqrt{\kappa} \rfloor$ is permitted. The number of included zeros then tends to infinity, the cap has a finite nonzero limit per zero, and the late imaginary-face flux tends to zero per zero. None of these statements asserts that the unnormalized flux tends to zero for every growing population.

The Vandermonde is essential for identifying the pair contribution, but it is not the whole physical cap. In this regime it has no leading phase increment at all. The dominant common phase comes from the canonical analytic factor. Keeping both gives a short proof and avoids trying to estimate two large cancelling force sums separately.

1.2 Relation to previous work and scope of novelty

The heat deformation belongs to the classical program of de Bruijn and Newman [1, 2]; Newman and Wu [3] describe its connections with statistical physics. Ki, Kim and Lee [4] proved eventual reality and simplicity at large height for every fixed positive heat time. Polymath [5], especially Theorem 1.5 and Section 9, obtained effective high-height approximations and zero counts. Those results use a different limiting regime from (1.6). We prove the needed complex-time estimates directly from the complete positive source.

Root dynamics and pair interaction are also established territory. Csordas, Smith and Varga [6] used zero-motion equations and inverse-square sums in their work on Lehmer pairs. Rodgers and Tao [7], Section 4 and Lemma 4.18, developed localized logarithmic Hamiltonians with exterior interaction terms in proving $\Lambda \geq 0$. Dobner [8] gives a different route to Newman's conjecture and its extension. Hall and Ho [9], Proposition 3.1, place polynomial heat-root motion in the Calogero–Moser setting. None of the finite pair identities used here is proposed as a new dynamics. Montgomery's pair-correlation theorem [10] concerns a different statistical limit; weak pair convergence alone does not supply the complex phase or uniform small-gap information needed by a physical cap.

The topological ingredients are classical as well. Whitehead's integral formula [11] and Chern–Simons transgression [12] explain the boundary form in (1.1). Rudolph [13] explicitly discusses reducing Hopf integrals by integration along fibers; therefore the general idea of reducing a three-form to a one-form is not new. The discriminant and resultant identities belong to classical elimination theory [14]. Papanicolaou, Kallitsi and Smyrlis [15], Theorem 4.1, prove that multiple zeros of an entire caloric function are isolated in complex spacetime. The local Hermite splitting used below is consistent with that result and with the real zero-number theory of Angenent [16], after time reversal. Pashaev and Gurkan [17] already connect Chern–Simons vortices, holomorphic linear evolution and zero dynamics; their gauge-field construction is distinct from the normalized first-jet boundary considered here.

Accordingly, the contribution proposed for assessment is the explicit physical boundary formula together with its population-uniform asymptotic estimates and complete closure accounting. The

elementary $O(m\eta)$ Vandermonde logarithm estimate in Lemma 3.1 and the narrow-strip source estimate are the quantitative steps that make the growing-population statement possible. We do not claim priority for the invariant, the discriminant identity, root-following coordinates, or heat-zero creation. A targeted primary-source review found the substantial overlaps just described; it does not establish that no equivalent quantitative theorem exists elsewhere. This is a research working manuscript, with originality of that narrower contribution still subject to specialist assessment.

2 The topological balance and the moving wall

We orient $\mathbb{C}^2$ by $dt \wedge ds \wedge dx \wedge dy$. All boundaries below are piecewise smooth, with the outward orientation. Corners have zero three-dimensional measure.

Lemma 2.1 (Local heat charge)**.** *Suppose $H \not\equiv 0$ is holomorphic near a point $p = (\tau_*, z_*)$, satisfies $H_\tau = -H_{zz}$, and $H(\tau_*, \cdot)$ has a zero of order $n \geq 2$ at z_*. Then the joint zero of (H, H_z) is isolated, and its local complex degree is $\binom{n}{2}$.*

Proof. Translate p to zero and write the leading spatial coefficient as $a \neq 0$. The convergent heat expansion has weighted leading term

$$H(\tau, z) = aP_n(\tau, z) + \text{terms of weighted degree greater than } n, \qquad P_n = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{n!(-\tau)^j z^{n-2j}}{j!(n-2j)!},$$

where τ has weight two and z weight one. The polynomial $P_n(1, z)$ is a scaled Hermite polynomial and has n distinct real roots. One elementary proof uses its orthogonality with a positive Gaussian weight: fewer than n distinct sign changes would produce a polynomial of degree below n whose product with P_n has one sign, contradicting orthogonality.

Put $\tau = v^2$ and $z = vw$. The quotient $v^{-n}H(v^2, vw)$ extends holomorphically to $v = 0$ and equals $aP_n(1, w)$ there. The implicit function theorem gives all n roots in the cluster as $z_j(v^2) = v(r_j + O(v))$, with distinct r_j. Consequently its monic Weierstrass discriminant is

$$\Delta(\tau) = c\tau^{n(n-1)/2}(1 + o(1)), \qquad c \neq 0.$$

It is holomorphic in τ, so its zero has the indicated integer order and is isolated. This also isolates the joint zero.

For completeness, the local intersection number of $H = 0$ and $H_z = 0$ equals the vanishing order of this discriminant. Indeed, Weierstrass preparation writes $H = UP$, with U a unit and P monic of degree n. Modulo P, $H_z = UP_z$; multiplication by the unit changes the resultant by a nonvanishing factor. The resultant of P and P_z is its discriminant up to sign. Equivalently, the determinant of multiplication by P_z on the free $\mathbb{C}\{\tau\}$-module $\mathbb{C}\{\tau\}[z]/(P)$ has this vanishing order, equal to the dimension of the local intersection algebra. A small regular perturbation counts that many positive holomorphic intersections, hence gives the local degree. □

At real coefficients the same expansion shows that, just after a real event, all n local roots are simple and real. Just before it there are n mod 2 real roots. Thus the real birth weight and complex degree stated in Section 1 follow. Positivity of the degree is not a sign estimate on each individual boundary face.

Lemma 2.2 (Complete boundary integral)**.** *Let W be a compact domain on whose boundary G is nonzero. Then*

$$\int_{\partial W} \mathcal{C} = \deg(G/|G|\,; \partial W) = \sum_{p \in W\colon G(p)=0} \binom{m_p}{2}. \tag{2.1}$$

Proof. On the unit sphere in $\mathbb{C}^2$, the form $\alpha = \mathrm{Im}(\bar{\zeta}_1 d\zeta_1 + \bar{\zeta}_2 d\zeta_2)$ satisfies $\alpha \wedge d\alpha = 2\, d\mathrm{Vol}_{S^3}$. Its integral is $4\pi^2$. This follows by evaluating at $(1,0)$ on $(\partial_{y_1}, \partial_{x_2}, \partial_{y_2})$ and using unitary invariance. The form $\mathcal{C}$ is the pullback of this normalized volume. Excise disjoint balls around the finitely many joint zeros and apply degree additivity, then Lemma 2.1. □

2.1 The exact cap integral

A regular projective root tube over a complex-time rectangle D is a holomorphic inverse

$$Z : D \times \{|q| \le \epsilon\} \longrightarrow \mathbb{C}, \qquad H_z(\tau, Z) \neq 0, \qquad \frac{H(\tau, Z)}{H_z(\tau, Z)} = q, \tag{2.2}$$

defined on a neighborhood of the displayed compact set and injective in each q fiber. The radius is constant on this whole tube. Differentiating the inverse equation gives $Z_q \neq 0$, so its parameterization preserves the complex spatial orientation. Its center $z(\tau) = Z(\tau, 0)$ follows a simple zero.

Proposition 2.3 (Phase reduction and zero side flux)**.** *For m disjoint tubes of the form (2.2), with a common radius ϵ, define*

$$h_j(\tau) = H_z(\tau, z_j(\tau)), \qquad \mathscr{D}(\tau) = \prod_{j=1}^{m} h_j(\tau). \tag{2.3}$$

For an oriented heat edge e, the integral over its root disks is

$$I(e) = \frac{k_\epsilon}{2\pi} \,\mathrm{Im} \int_e d \log \mathscr{D}. \tag{2.4}$$

Every wall $|q| = \epsilon$ has identically zero pullback of $\mathcal{C}$. Furthermore,

$$\mathscr{D} = (-1)^{m(m-1)/2} \Delta \mathscr{U}. \tag{2.5}$$

The factor $U = H/\prod_j (z - z_j)$ need only be a nonvanishing holomorphic factor near the selected roots. It can have zeros outside those neighborhoods; no infinite product is required.

Proof. In one tube write $h(\tau, q) = H_z(\tau, Z(\tau, q))$ and $\phi = \mathrm{Im} \log h$, using a continuous logarithm on the contractible parameter domain. The physical jet is exactly $h(q, 1)$, so

$$\begin{aligned} \mathcal{A} &= d\phi + a(q), & a(q) &= \frac{\mathrm{Im}(\bar{q}\, dq)}{1 + |q|^2}, \\ d\mathcal{A} &= \omega(q), & \omega(q) &= \frac{2\, du \wedge dv}{(1 + u^2 + v^2)^2}, \quad q = u + iv. \end{aligned} \tag{2.6}$$

Since $a \wedge \omega = 0$, $\mathcal{C} = d\phi \wedge \omega/(4\pi^2)$. The derivative of ϕ in the edge direction is harmonic in q. Its weighted average over the disk is its center value times

$$\int_{|q| \le \epsilon} \omega = \int_0^{2\pi} \int_0^{\epsilon} \frac{2r}{(1 + r^2)^2}\, dr\, d\vartheta = 2\pi k_\epsilon.$$

This proves (2.4) after summing the tubes. On the wall $q = \epsilon e^{i\vartheta}$, the two-form ω pulls back to zero, proving the side assertion.

At a selected simple root, $H_z(\tau, z_j) = U(\tau, z_j) \prod_{i \neq j} (z_j - z_i)$. Multiplying these identities pairs the factors belonging to every unordered pair and gives (2.5). □

The vanishing side flux is a geometric statement, not a bound on the speed of a root. It remains true when the disks move rapidly. A radius allowed to vary *along* the tube would produce $dr \wedge d\vartheta$ terms and would not have the same property. We allow a different constant radius for each chosen terminal T.

Theorem 2.4 (The complete transport formula)**.** *Suppose W is an event-containing compact domain with $G \neq 0$ on its boundary, whose terminal face is the union of m disks from Proposition 2.3. Decompose its other boundary into the initial face, the two imaginary-time faces, regular projective walls, and all remaining spatial or joining faces. Their oriented integrals are $-\mathcal{I}_0$, $\mathcal{J}_+$, $\mathcal{J}_-$, 0, and $\mathcal{R}$, respectively. Then (1.2) holds with (1.3).*

If $A(W)$ is the number of distinct real creation events in the interior, and separately proved bounds give

$$\mathcal{I}_0 \geq L_0, \quad \mathcal{I}_T \leq U_T, \quad \mathcal{J}_+ + \mathcal{J}_- \leq U_J, \quad \mathcal{R} \leq U_R,$$

then

$$A(W) \leq \mathcal{Q} \leq U_T + U_J + U_R - L_0. \tag{2.7}$$

Proof. Lemma 2.2 gives $\mathcal{Q} = \mathcal{I}_T - \mathcal{I}_0 + \mathcal{J}_+ + \mathcal{J}_- + \mathcal{R}$. Substitute Proposition 2.3. Each real event contributes at least one to the nonnegative sum defining $\mathcal{Q}$, proving the bound. □

This is a useful interface for estimates, but it does not prove that the four required one-sided bounds have favorable sizes. In particular, a bound for a closure expressed in terms of $\mathcal{Q}$ itself characterizes the closure; it cannot be inserted into (2.7) as independent information about $\mathcal{Q}$.

2.2 The other two faces of a regular tube

Assume real type, meaning $H(\bar{\tau}, \bar{z}) = \overline{H(\tau, z)}$, and let the root branches be real at real time on $[a, b]$. Each $h_j(t)$ then has a constant real sign. Choose

$$\Theta(t, Y) = \sum_j \operatorname{Im} \log \frac{h_j(t + iY)}{h_j(t)}, \tag{2.8}$$

where each logarithm is continued vertically from zero imaginary time. Conjugation and Proposition 2.3 give

$$\boxed{\mathcal{I}_t = \frac{k_\epsilon}{\pi}\Theta(t, Y), \qquad \mathcal{J}_+ = \mathcal{J}_- = \frac{k_\epsilon}{2\pi}\{\Theta(a, Y) - \Theta(b, Y)\} = \frac{\mathcal{I}_a - \mathcal{I}_b}{2}.} \tag{2.9}$$

The orientation on the upper imaginary face is $-dt \wedge du \wedge dv$; on the lower it is $dt \wedge du \wedge dv$. Thus the complete regular tube obeys $\mathcal{I}_b - \mathcal{I}_a + \mathcal{J}_+ + \mathcal{J}_- = 0$, as it must because it contains no joint zero. Attaching tubes to an event region does not remove the complement of their attachment disks.

3 A Vandermonde estimate with linear population cost

A termwise estimate of pair phases suggests an unnecessary harmonic sum. Summing the linear part first removes it. This observation is useful because a cluster has $m(m-1)/2$ pairs, whereas its physical cap should be compared with a population of size m.

Lemma 3.1 (Perturbing a clock discriminant)**.** *Let $c_j = c_1 + (j-1)\pi$, $1 \le j \le m$, and let $w_j = c_j + \delta_j$ with $|\delta_j| \le \eta \le \pi/8$. For $m = 1$ both discriminants below are one. Using the logarithm obtained by continuously moving the points from c_j to w_j, one has*

$$\left|\log \frac{\prod_{i<j}(w_j - w_i)^2}{\prod_{i<j}(c_j - c_i)^2}\right| \le \frac{4(m-1)}{\pi}\eta + \frac{8m}{9}\eta^2 \le 2m\eta. \tag{3.1}$$

Proof. Set $u_{ij} = (\delta_j - \delta_i)/(\pi(j-i))$, so $|u_{ij}| \le 1/4$. Each pair ratio stays in a disk about one avoiding zero. Hence its logarithm is unambiguously the power-series logarithm, and

$$\log(\Delta(w)/\Delta(c)) = 2\sum_{i<j} u_{ij} + 2\sum_{i<j}\{\mathrm{Log}(1+u_{ij}) - u_{ij}\}.$$

With $H_0 = 0$ and $H_j = \sum_{d=1}^{j} d^{-1}$,

$$\sum_{i<j}\frac{\delta_j - \delta_i}{j-i} = \sum_{j=1}^{m}\delta_j(H_{j-1} - H_{m-j}).$$

For each d, the two indicator sequences in $H_{j-1} - H_{m-j}$ differ at $2\min(d, m-d)$ indices. Therefore

$$\sum_{j=1}^{m}|H_{j-1} - H_{m-j}| \le \sum_{d=1}^{m-1}\frac{2\min(d, m-d)}{d} \le 2(m-1).$$

This bounds the linear part by $4(m-1)\eta/\pi$. The elementary remainder estimate $|\mathrm{Log}(1+u) - u| \le |u|^2/[2(1-|u|)]$ bounds the rest by

$$\frac{16\eta^2}{3\pi^2}\sum_{d=1}^{m-1}\frac{m-d}{d^2} \le \frac{8m}{9}\eta^2.$$

Finally $4/\pi + \pi/9 < 2$, proving the last inequality. □

The estimate controls a continuously lifted complex logarithm, not only the modulus of a determinant. A lower gap bound by itself would not give the same phase information. The proof is elementary; no optimality or priority for its numerical constants is asserted.

4 The complete theta source on a narrow spatial strip

We now prove Theorem 1.1. The central simplification is to work in disks around the clock points, whose imaginary parts are bounded independently of the rank. On such disks the exponential bound for sine and cosine involves $|\mathrm{Im}\, w|$, not $|w|$. The real width can therefore grow without introducing an $e^{\pi K}$ loss.

Lemma 4.1 (Weighted concentration)**.** *Let U be the positive random variable with density $e^{Tu^2}\Phi(u)/H(T,0)$ on $u \ge 0$, and put $V = U - L$. For each fixed $a \ge 0$ and $j = 1, 2, 3$,*

$$M_j(T,a) := \mathbb{E}\big[|V|^j e^{a|V|/L}\big] = O_a(\kappa^{-j/2}). \tag{4.1}$$

The constants do not depend on a root rank or a selected population.

Proof. All theta indices can be retained in the factorization

$$e^{Tu^2}\Phi(u) = 2\pi^2 e^{q_T(u)} b(u), \qquad q_T(u) = Tu^2 + 9u - \pi e^{4u},$$
$$b(u) = \sum_{n\geq 1} \left(n^4 - \frac{3n^2 e^{-4u}}{2\pi} \right) e^{-\pi(n^2-1)e^{4u}}.$$

For all $u \geq 0$,

$$0 < b_* := 1 - \frac{3}{2\pi} \leq b(u) \leq B_* := \frac{1}{1 - 16e^{-3\pi}}. \tag{4.2}$$

The lower bound uses the first term; all terms are positive. For the upper bound discard the subtraction and compare successive terms $n^4 e^{-\pi(n^2-1)}$ with a geometric series of ratio at most $16e^{-3\pi}$.

For large T, $L \to \infty$, $T/\kappa \to 0$, and

$$q_T(L+v) - q_T(L) = Tv^2 - \pi e^{4L}(e^{4v} - 1 - 4v).$$

On $v \geq -1/4$ one has $q_T''(L+v) \leq -\kappa/(2e)$ eventually. Hence this difference is at most $-c\kappa v^2$, where $c = 1/(4e)$. On $0 \leq u \leq L - 1/4$, q_T has at most one local minimum and no local maximum: q_T' first increases and then decreases, with its later zero at L. Since $q_T(L - 1/4) > q_T(0)$ eventually, the maximum on that interval is its right endpoint. Thus the entire far-left part has exponent at most $q_T(L) - c\kappa/16$.

On $|v| \leq \kappa^{-1/2}$, Taylor's formula gives $q_T(L+v) - q_T(L) \geq -1/2 - O(\kappa^{-1/2})$. The normalization is therefore at least a positive constant times $e^{q_T(L)}/\sqrt{\kappa}$. On the near and right parts put $w = \sqrt{\kappa} v$. Dividing by this lower normalization bounds the moment by a constant times

$$\kappa^{-j/2} \int_{\mathbb{R}} |w|^j e^{-cw^2 + a|w|/(L\sqrt{\kappa})}\, dw + L^{j+1}\sqrt{\kappa}\, e^{a - c\kappa/16}.$$

The integral is bounded for large T; the second term is smaller than every negative power of κ. This proves the assertion. Both tails, the normalization, and all source indices were included. □

Lemma 4.2 (A rank-uniform complex-time approximation)**.** *For real $|s| \leq Y$ define*

$$F(s,w) = \frac{H(T + is, w/L)}{H(T,0)}, \qquad c_j(w) = \partial_w^j \cos w.$$

On $|\operatorname{Im} w| \leq a$ and $|w| \leq Z$, for $j = 0, 1, 2, 3$,

$$|\partial_w^j F - e^{isL^2} c_j(w)| \leq e^a \left\{ \sum_{k=1}^{j} \binom{j}{k} \frac{M_k(T,a)}{L^k} + \frac{Z M_1(T,a)}{L} + Y\big(2LM_1(T,a) + M_2(T,a)\big) \right\}. \tag{4.3}$$

The sum is zero for $j = 0$. In particular, for fixed a, Y and $Z \leq \pi K + 1$, the right side is $O_{a,Y}(\varepsilon_T)$.

Proof. Put $R = U/L$. Differentiation under the convergent complete source gives $\partial_w^j F = \mathbb{E}[R^j e^{isU^2} c_j(Rw)]$. Split the error into the changes in R^j, in $c_j(Rw)$, and in e^{isU^2}. The first is bounded using

$$|R^j - 1| \leq \sum_{k=1}^{j} \binom{j}{k} |V|^k / L^k, \qquad |c_j(Rw)| \leq e^a e^{a|V|/L}.$$

The line segment from w to Rw gives $|c_j(Rw) - c_j(w)| \leq Z|V| e^a e^{a|V|/L}/L$. Finally, because s, U, L are real,

$$|e^{isU^2} - e^{isL^2}| \leq |s|\,|U^2 - L^2| \leq Y(2L|V| + V^2).$$

Taking expectations proves (4.3). No complex-time derivative of an asymptotic error has been taken. □

4.1 Actual moving disks and their rank labels

Let $s_j = (j - 1/2)\pi$, $r_1 = 1/3$, and $\rho = 1/4$. For $|v| \le r_1$ consider

$$F(s, w) - vF_w(s, w) = 0, \qquad w_{0,j}(v) = s_j + \arctan v. \tag{4.4}$$

The principal branch of arctan is analytic here. On $|w - w_{0,j}(v)| = \rho$, the reference equation is

$$e^{isL^2}(\cos w + v \sin w) = -\sin(s_j)e^{isL^2}\sqrt{1 + v^2}\,\sin(w - w_{0,j}(v)).$$

Its modulus is at least $\sqrt{1 - r_1^2}(2\rho - \sinh\rho) > 0$, uniformly in j. These disks have $|\operatorname{Im} w| \le \operatorname{arctanh}(1/3) + 1/4 < 1$ and $|w| \le \pi K + 1$ for $j \le K$. Lemma 4.2 and Rouché's theorem therefore give exactly one simple solution in each disk, once ε_T is small. Throughout the disk,

$$|\sin w| = |\cos(w - s_j)| \ge 2 - \cosh\big(\operatorname{arctanh}(1/3) + 1/4\big) > 0.$$

Thus $F_w \ne 0$ at the actual solutions as well. The solutions depend holomorphically on $\tau = T + is$ and v locally, and uniqueness patches them along the compact strip. Since $H/H_z = v/L$, they are the required inverse disks in the physical coordinate $q = v/L$. Injectivity in v follows by recovering v from that quotient. Their images are disjoint, since the disks around adjacent clock centers have total diameter less than π.

At $s = 0$ the source and the reference are real. Choose any fixed $0 < \beta < \pi/4$. On the real interval from zero to $s_K + \beta$, the uniform errors for F and F_w tend to zero. On each $[s_j - \beta, s_j + \beta]$ the derivative has constant nonzero sign and the two endpoint values have opposite signs. On the remaining intervals $|\cos w| \ge \sin\beta$, so no zero is missed. This identifies the roots as exactly the first K positive real roots, not merely K roots near guessed labels.

At the centers $v = 0$, the same estimates imply, uniformly for $j \le K$ and $|s| \le Y$,

$$Lz_j(T + is) = s_j + \delta_j(s), \qquad |\delta_j(s)| \le C_Y\varepsilon_T, \tag{4.5}$$

$$h_j(T + is) = H(T, 0)L[-\sin(s_j)]e^{isL^2}(1 + \rho_j(s)), \qquad |\rho_j(s)| \le C_Y\varepsilon_T. \tag{4.6}$$

For (4.5), use the lower linear bound for sin on the fixed root disk. For (4.6), evaluate the F_w estimate there and use the bounded derivative of sin. Increasing C_Y if needed pays both bounds; it is independent of the population. The ratio in (4.6) remains in a disk about one of radius less than $1/2$, so it has a single near-one logarithm throughout the cap. This is what controls a lifted phase, even though YL^2 itself becomes large.

4.2 Proof of the terminal estimates

Equation (4.6), summed over the selected block, gives

$$\Theta(T, Y) = mYL^2 + O_Y(m\varepsilon_T). \tag{4.7}$$

Substitution in (2.9), with $k_\epsilon = r^2/(L^2 + r^2)$, proves (1.7). This argument uses endpoint logarithms and the exact disk integral; it does not multiply an uncontrolled derivative error by a growing cap volume.

Apply Lemma 3.1 to the unfolded roots in (4.5). The reference discriminant is positive real and independent of s; the factor $L^{-m(m-1)}$ is also positive real. For $\eta = C_Y\varepsilon_T \le \pi/8$ the two endpoint logarithms have modulus at most $2m\eta$ each. Therefore

$$\left|\frac{k_\epsilon}{2\pi}\operatorname{Im}\int_{T-iY}^{T+iY} d\log\Delta\right| \le \frac{2k_\epsilon m\eta}{\pi}. \tag{4.8}$$

This proves (1.8). The factorization (2.5) proves the corresponding analytic-factor estimate by subtraction. In particular,

$$\frac{k_\epsilon}{2\pi m}\,\mathrm{Im}\int_{T-iY}^{T+iY} d\log\mathscr{U} = \frac{r^2 Y L^2}{\pi(L^2+r^2)} + O_Y\left(\frac{r^2\varepsilon_T}{L^2}\right). \tag{4.9}$$

If $Y = Y_T = S/L^2$ with fixed $S > 0$, repeat (4.3) with that value, rather than using a fixed-width error. Its scale becomes $O_S((K+1)/(L\sqrt{\kappa}))$. The same construction gives

$$\frac{\mathcal{I}_{T,m}}{m} = \frac{r^2 S}{\pi(L^2+r^2)} + O_S\left(\frac{r^2(K+1)}{L^3\sqrt{\kappa}}\right) = O_{r,S}(L^{-2}). \tag{4.10}$$

This is a uniform estimate per root even when m grows. An absolute vanishing cap follows, for example, if also $m = o(L^2)$.

4.3 Proof of the late edge estimate

On $[\theta T, T]$, $L_t/L_T \to 1$ and κ_t/κ_T stays bounded above and below by positive constants depending on θ. The preceding error estimates are therefore uniform there. For the fixed physical radius $\epsilon = r/L_T$, the local inverse variable is $v = L_t q$, with $|v| \le r$. The same construction gives disjoint holomorphic tubes over the whole rectangle: the unique inverse branches agree on overlaps. No assertion about continuation back to time zero is needed.

At each endpoint $b \in \{\theta T, T\}$, formula (4.7) reads $\mathcal{I}_b/m = k_\epsilon Y L_b^2/\pi + O_{Y,\theta}(k_\epsilon\varepsilon_T)$. Thus (2.9) yields the more precise finite asymptotic

$$\frac{\mathcal{J}_\pm}{m} = \frac{r^2 Y(L_{\theta T}^2 - L_T^2)}{2\pi(L_T^2+r^2)} + O_{Y,\theta}\left(\frac{r^2\varepsilon_T}{L_T^2}\right). \tag{4.11}$$

Differentiating the defining saddle equation gives the exact identity $L_t' = 2L_t/\kappa_t \sim 1/(4t)$, uniformly on this interval. Hence $L_T - L_{\theta T} = \frac{1}{4}\log(1/\theta) + o(1)$. Substitution proves (1.9) and completes Theorem 1.1.

These tubes contain no collision. Their late edge estimate is a proved bound on a named part of a longer boundary, not an estimate on the earlier connection to an event-containing region. The latter has a different asymptotic behavior, described next.

5 What remains when a boundary encloses an event

An estimate on a late regular tube does not determine the flux on a complete boundary starting at time zero. We now give a quantitative description of this difference. It also shows why the extra terms in (1.2) cannot be suppressed by taking a thinner strip.

Let a fixed core be

$$W_c(Y) = [0,a] + i[-Y,Y] \;\times\; \{|z - x_*| \le R\}, \qquad x_* \in \mathbb{R}, \quad 0 < Y \le Y_0.$$

Assume real type, a nonzero jet on its boundary for these Y, and that the only joint zeros in the core are a fixed finite set of real events in its interior. Denote their total complex charge by Q. The initial face, the whole face $t = a$, and the spatial wall are uniformly separated from the joint zeros. Suppose on their union for $|s| \le Y_0$ that

$$|G| \ge b > 0, \qquad \|DG\|_{\mathrm{op}} \le D. \tag{5.1}$$

These are compact boundary quantities, not assumptions about a floor at the events. Assume m disjoint real root tubes attach along disks of the $t = a$ face and continue to the terminal cap. The complement of those disks remains part of the boundary.

For the canonical flow, a fixed finite set of real simple branches that remains simple for $t \geq a$ admits such tubes for every finite T. A compact initial segment has a common positive complex-time width and inverse radius by the implicit function theorem. The late part is covered by Section 4. Combining them, and decreasing the common width if necessary, permits a fixed small $Y > 0$ and radius r/L_T for all sufficiently large T. For example, branches starting at $a > 1/2$ stay simple: de Bruijn's reality theorem and the local splitting lemma rule out a later collision. This continuation statement neither asserts an actual earlier theta event nor supplies a height-uniform core floor.

Theorem 5.1 (Quantitative closure allocation)**.** *In this fixed-core geometry let $\mathcal{R}_T^{\mathrm{all}}$ be the signed sum of the initial, joining and local spatial faces, including the initial sign $-\mathcal{I}_0$. Set*

$$B = \frac{D^3}{2\pi^2 b^3}(4\pi R^2 + 4\pi R a). \tag{5.2}$$

Every projective tube wall has zero flux, and

$$|\mathcal{R}_T^{\mathrm{all}}| \leq BY, \qquad \boxed{\mathcal{J}_+ = \mathcal{J}_- = \frac{Q - \mathcal{I}_{T,m} - \mathcal{R}_T^{\mathrm{all}}}{2}.} \tag{5.3}$$

For the following asymptotic assertions, assume that H is the canonical flow (1.4) and that the attached roots have fixed positive ranks. If $Y_T = S/L_T^2$ and $\epsilon_T = r/L_T$, with fixed S, r and fixed attached population, then

$$\mathcal{I}_{T,m} = O(mL_T^{-2}), \qquad \left|\mathcal{J}_\pm - \frac{Q}{2}\right| \leq \frac{BY_T + |\mathcal{I}_{T,m}|}{2} = O((B+m)L_T^{-2}). \tag{5.4}$$

For fixed admissible $Y > 0$, let $R_c(Y)$ be the signed sum of the three non-imaginary faces of the unmodified core. Then

$$\begin{aligned} \mathcal{I}_{T,m} &\longrightarrow \frac{mr^2 Y}{\pi}, \\ \mathcal{R}_T^{\mathrm{all}} &\longrightarrow R_c(Y), \qquad\qquad |R_c(Y)| \leq BY, \\ \mathcal{J}_\pm &\longrightarrow \frac{Q - R_c(Y)}{2} - \frac{mr^2 Y}{2\pi}. \end{aligned} \tag{5.5}$$

The errors in these fixed-width limits are $O(L_T^{-2}) + O(\varepsilon_T/L_T^2)$ with fixed core and population constants.

Proof. The normalized sphere-volume expression for (1.1) gives a comass bound $D^3/(2\pi^2 b^3)$ on the faces in (5.1). Their total three-volume is at most $Y(4\pi R^2 + 4\pi R a)$. Removing the attachment disks only reduces this upper bound. This proves the first inequality.

Simultaneous conjugation $(t, s, x, y) \mapsto (t, -s, x, -y)$ preserves the ambient orientation, exchanges the outward imaginary faces, and sends $\mathcal{A}$ and $d\mathcal{A}$ to their negatives. It therefore preserves $\mathcal{C}$ and proves $\mathcal{J}_+ = \mathcal{J}_-$. The complete boundary has charge Q by Lemma 2.2, yielding (5.3). Equation (4.10) proves (5.4).

For fixed width, the removed attachment caps at $t = a$ have size $O(k_{\epsilon_T}) = O(L_T^{-2})$, because their center derivatives are nonvanishing on a fixed compact segment and their lifted phase increments are bounded. Thus $\mathcal{R}_T^{\mathrm{all}} = R_c(Y) + O(L_T^{-2})$. The terminal estimate (1.7) and the exact balance give (5.5) with the stated errors. □

The half-charge limit is a characterization of the remaining flux, not an independent upper bound on the event count. Its constants belong to a fixed core. Allowing height, multiplicity or the number of cores to grow requires new estimates for b, D, R, a; none is implicit here. Shrinking Y also changes the initial cap being integrated. At fixed Y, the initial core and its initial integral remain fixed, and the nonzero limits in (5.5) give the appropriate comparison.

5.1 Two exact controls

The quadratic heat solution $H(\tau, z) = z^2 - 2\tau$ has charge one at the origin. Its inverse root disks are

$$Z_{\pm}(\tau, q) = q \pm \sqrt{q^2 + 2\tau}.$$

For $t \geq a > 0$ and $\epsilon^2 < 2a$, these are disjoint regular tubes on a sufficiently small imaginary-time strip. Their two-root terminal cap is exactly

$$\mathcal{I}_T = \frac{k_\epsilon}{\pi} \arctan(Y/T). \tag{5.6}$$

Indeed $\mathscr{D} = -8\tau$, so the vertical phase increment is $2\arctan(Y/T)$. The cap tends to zero and the sides have zero flux, although an attached core still has charge one. The two inverse branches pinch at $q^2 = -2\tau$ when continued backward; one cannot extend a fixed-radius regular disk bundle through the event. This example checks the need for a joining region.

For $\lambda > 0$, the exact heat solution $H(\tau, z) = e^{\lambda^2 \tau} \sin(\lambda z)/\lambda$ has stationary simple roots. On any fixed finite collection of their projective disks, Δ is constant but $\mathscr{D}$ has phase rate $m\lambda^2$. Hence

$$\mathcal{I}_T = \frac{k_\epsilon m \lambda^2 Y}{\pi}. \tag{5.7}$$

This cap is nonzero; the full regular tube still has zero charge. It checks that root positions alone do not determine the physical cap. The leading phase in Theorem 1.1 is the canonical, increasing-frequency counterpart of this calculation.

Both controls satisfy the heat equation but lack the unchanged theta source. Conversely, the source proof uses only the displayed positive density bounds and concentration: it is not claimed to distinguish theta from every other source with the same asymptotic properties. No collision exclusion can be inferred merely from the late clock pattern.

6 What the formula now makes accessible

The initial-to-future identity (1.2) and its independent-bound consumer (2.7) are the organizing framework. The estimates proved here supply a terminal cap, its Vandermonde and analytic-factor decomposition, exactly vanishing projective sides, and relative late-edge asymptotics for a growing population. They also identify a substantial surviving term at a fixed event core. These are concrete bounds with different domains of validity; their domains must match before they can be combined to bound events.

The next estimate needed for a full time-zero application is on the complete earlier attachment and imaginary-time boundary, with constants controlled as the spatial height grows. A meaningful first target is a bound for that contribution divided by the enclosed zero population. It need not be small enough to exclude every event. One could then use an independently obtained lower bound on the fixed initial cap in (2.7). Neither that lower bound nor a uniform early-boundary estimate is

proved here. The fixed-core half-charge formula cannot replace them, because it already contains the unknown charge.

There is also a separate real-variable route. At fixed positive heat time, the high-height estimates of Polymath yield zero-free moving barriers and finite budgets for later real births. Those budgets do not estimate the physical imaginary-time integrals in this paper. Likewise a pair-correlation law for real zeros does not supply the analytic factor in (1.2). Keeping these distinctions explicit leaves a modest but useful conclusion: the topological formula has quantitatively tractable pieces, and the remaining task is located on a specified boundary rather than hidden in a presumed vanishing cap.

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All asymptotic statements have analytic proofs. No numerical onset, theta-zero certificate, priority certification, proof of RH, or proof of initial simplicity is asserted. The accompanying literature and proof review records source access, overlap, control examples and the remaining novelty question.

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